Attention-Based Estimation of Neural Entropy

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Abstract

Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience.

1 Introduction

Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons.

2 Related Work

2.1 Entropy estimators

We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered.

2.2 Attention models

Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons.

3 Method

The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions.

3.1 Architecture

Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week.

4 Experiments

These observations agree with prior reports from independent groups. Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions.

Model	BLEU	Params
1 Baseline	24.1	60M
2 Attention	28.4	65M
3 Ours	29.0	64M

Table 1: Comparison with prior models on the benchmark.

4.1 Results

Alternative explanations based on arousal alone were considered. The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits

separated by one week. The analysis controlled for head motion and physiological noise. Higher entropy was associated with broader subjective experience. These observations agree with prior reports from independent groups.

5 Conclusion

The effect remained after correction for multiple comparisons. Brain entropy describes the diversity of neural states visited over time. We measured signal complexity with established estimators across sessions. Each participant completed two scanning visits separated by one week. The analysis controlled for head motion and physiological noise.

References

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A Proofs

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